
Exercise 7 – Department Salary Report

Workflow

```
create_department_file
    ↓
calculate_department_salary
    ↓
generate_department_report
```

Input

```
/tmp/departments.txt
```

Content

```
IT,45000
HR,35000
Finance,50000
IT,55000
Finance,40000
HR,30000
```

Task 2

Calculate total salary for each department.

Expected

```
IT = 100000
HR = 65000
Finance = 90000
```

Task 3

Generate

/tmp/department_report.txt

Exercise 8 – Electricity Bill Summary

Workflow

```
create_bill_file
    ↓
calculate_total_units
    ↓
generate_bill_summary
```

Input

/tmp/electricity.txt

Content

```
Rahul,210
Priya,180
Amit,300
Sneha,150
Kiran,260
```

Task 2

Calculate

- Total Units
- Average Units

Task 3

Generate

/tmp/bill_summary.txt

Example

```
Customers = 5
Total Units = 1100
Average Units = 220
```

Exercise 9 – Exam Result Report

Workflow

```
create_result_file
    ↓
count_pass_fail
    ↓
generate_result_summary
```

Input

/tmp/results.txt

Content

```
Rahul, Pass
Priya, Fail
Amit, Pass
Sneha, Pass
Kiran, Fail
Megha, Pass
```

Task 2

Calculate

- Total Pass
- Total Fail

Task 3

Generate

/tmp/result_summary.txt

Exercise 10 – Online Orders

Workflow

```
create_orders
    ↓
calculate_order_value
    ↓
generate_sales_report
```

Input

orders.csv

```
product,quantity,price
Laptop,1,70000
Mouse,4,500
Monitor,2,12000
Keyboard,3,1500
```

Task 2

Calculate revenue for each product.

Task 3

Generate

/tmp/sales_report.txt

Include

```
Total Revenue
Highest Selling Product
```

Exercise 11 – Temperature Analysis

Workflow

```
create_temperature_file
    ↓
find_highest_temperature
    ↓
generate_weather_report
```

Input

/tmp/temperature.txt

Content

```
Monday, 34
Tuesday, 36
Wednesday, 31
Thursday, 38
Friday, 35
Saturday, 33
Sunday, 32
```

Task 2

Calculate

- Highest Temperature
- Average Temperature

Task 3

Generate

/tmp/weather_report.txt

Exercise 12 – Bank Transaction Summary

Workflow

```
create_transactions
    ↓
calculate_balance
    ↓
generate_account_report
```

Input

/tmp/transactions.txt

Content

```
Deposit,10000
Withdraw,2500
Deposit,4000
Withdraw,1500
Deposit,2000
```

Task 2

Calculate

- Total Deposit
- Total Withdrawal
- Final Balance

Task 3

Generate

/tmp/account_report.txt

Exercise 13 – Employee Age Report

Workflow

```
create_employee_file
    ↓
calculate_average_age
    ↓
generate_age_report
```

Input

/tmp/employees.txt

Content

```
Rahul,28
Priya,31
Amit,42
Sneha,26
Kiran,38
```

Task 2

Calculate

- Youngest Employee
- Oldest Employee
- Average Age

Task 3

Generate

/tmp/age_report.txt

Exercise 14 – Course Enrollment Report

Workflow

```
create_enrollment_file
```

↓

```
count_students
```

↓

```
generate_course_report
```

Input

```
/tmp/enrollments.txt
```

Content

```
Python,Rahul  
Python,Priya  
SQL,Amit  
Python,Sneha  
Power BI,Kiran  
SQL,Megha  
Power BI,Arjun
```

Task 2

Count students enrolled in each course.

Expected

```
Python = 3  
SQL = 2  
Power BI = 2
```

Task 3

Generate

```
/tmp/course_report.txt
```